

and thousands of engineers every year. However, very few of these graduates are actually ready for the industry. In fact, according to studies done by McKinsey and Nasscom, only one-fourth of the graduates are ready for the industry," informs Sabih Ahmad Kidwai, head-human resources (HR), Rockwell Automation India.

Automation industry is very niche. The industry itself is in a nascent stage in India and so we don't have a huge talent pool available for the industry. On top of it, there are very few colleges that actually teach automation effectively.

Roles available

"Most automation tasks require the engineer to be on-site, which means a lot of travelling and outdoor work. Those who prefer a comfortable environment—sitting all day in front of a computer in a controlled temperature—will definitely not like the automation field and should not think about automation as a career," says Raj Singh Rathee, managing director, Kuka Robotics (India).

As for opportunities available, one can get into project, application, design or sales engineering. Then there are roles for industry experts. Of course, these are not for freshers but people who have been in the industry over a period of time. For example, there are power generation experts who know a power generation unit's in and out. They know where automation is working, how automation would save energy or increase speed, and so on.

The main metric is grip on control. And as a design professional, you are expected to acquire the knowledge and skills needed to design practical control loop as per that metric. If you can achieve these, instrumentation and maintenance related solutions will be automatically on your fingertips. To make maximum use of the situation, you need to have expertise to survive in the global-quality working environment of automation.

Automation is a very technical field and therefore domain expertise and knowledge of processes is very important. Because of the technology-driven

Why choose automation

The opportunities in industrial automation are widening due to:

1. The increase in the number of companies going in for automation
2. Better product quality expectation
3. Higher level of automation in the companies
4. More areas getting integrated in automation
5. Increase in the productivity

—Raj Singh Rathee, managing director, Kuka Robotics (India)

nature of this field, professionals must at all times keep themselves abreast of the latest improvements and technological upgradations that are taking place, instead of narrowly focusing on their departmental concerns.

Money matters

For a fresh engineering graduate, the salary could start from Rs 350,000-400,000 per annum. It could go well up to Rs 600,000 per annum for graduates from engineering institutes like IITs. However, the scene is not so encouraging for diploma holders.

You could start at a junior level as part of a major project and grow to become a project leader in 10-12 years' time frame. In most cases, salaries are proportional to the cost of the projects. Notably, the salary is on the higher side for design section.

What's expected from candidates

Candidates willing to take up a career in automation should have electrical, electronics and communication, instrumentation and control, or computer engineering background. They are expected to have an orientation for fundamentals of automation. It helps if the candidate has done some work in automation lab of his college.

"Courses available in colleges are bachelor's degree in different areas of technology such as electronics and communication, computers, instrumentation and control engineering, electrical engineering, information technology and me-

chanical engineering. In addition, there are courses available at the postgraduate level in different specialisations," informs Dr Alok Prakash Mittal, professor and head, Division of Instrumentation & Control Engineering, Netaji Subhas Institute of Technology.

Recruiters also look for the learning aptitude in candidates. The selected candidates will have to spend initial few years to move around and learn. This is unlike the IT industry where engineers sit in an air-conditioned office in a fancy building. Only those who are willing to fold their sleeves, hop on to a bus and go to plants at far-flung places around the country to learn and deliver are going to make it big.

"Anyone who has to grow in this field must have a field experience. So experience is more important than qualification for higher levels," says Rathee.

Rockwell Automation does campus selection. "We don't look for technology prowess, we just want the candidates to be aware of automation fundamentals—like what is automation, what are the tools and components, how are these connected, and how does a programmable logic controller (PLC) or supervisory control and data acquisition (SCADA) work. We look for people who are agile to the environment and have some curricular engagement. Thus we look for a holistic personality. We prefer candidates who are well-read and aware, and who have opinion on the environment around us—whether it's technical, political or social," informs Kidwai.

Once a candidate is selected, "We provide extensive on-job training, coaching and counselling. Apart from that, there is a lot of classroom or hands-on training. Normally, the programme is one-year long—three months in the classroom and nine months on the job," adds Kidwai.

Fundamentals go a long way

You need to understand automation elements from both the theoretical and practical point of view.

Automation is a step beyond